

## ***Nanomaterials in consumer products: a challenging analytical problem.***

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Many products used in everyday life are made with the assistance of nanotechnologies. Cosmetic, pharmaceuticals, sunscreen, powdered food are only few examples of end products containing nano-sized particles (NPs), generally added to improve the product quality. To evaluate correctly benefits vs. risks of engineered nanomaterials and consequently to legislate in favor of consumer's protection, it is necessary to know the hazards connected with the exposure levels. This information implies transversal studies and a number of different competences. On analytical point of view the identification, quantification and characterization of NPs in food matrices and in cosmetic or personal care products pose significant challenges, because NPs are usually present at low concentration levels and the matrices, in which they are dispersed, are complexes and often incompatible with analytical instruments that would be required for their detection and characterization. This paper focused on some analytical techniques suitable for the detection, characterization and quantification of NPs in food and cosmetics products, reports their recent application in characterizing specific metal and metal-oxide NPs in these two important industrial and market sectors. The need of a characterization of the NPs as much as possible complete, matching complementary information about different metrics, possible achieved through validate procedures, is what clearly emerges from this research. More work should be done to produce standardized materials and to set-up methodologies to determine number-based size distributions and to get quantitative data about the NPs in such a complex matrices.